

LAB EXPERIMENT ANALYSIS REPORT

Experiment 2: Introduction to Socket Programming Using Python

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Course: 23CSE302 - Computer Networks

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Date of Experiment: 22/06/2026

Maximum Marks: 10

TABLE OF CONTENTS

1. Executive Summary
2. Experiment Objectives
3. Technical Overview
4. Implementation Analysis
5. Code Walkthrough
6. Screenshot Analysis
7. Results & Inference
8. Lab Scenario Solution
9. Extension Task: Multi-threading
10. Key Learnings & Takeaways

1. EXECUTIVE SUMMARY

This experiment provides a comprehensive hands-on introduction to socket programming using Python, focusing on implementing basic client-server communication architecture. The experiment successfully demonstrates fundamental networking concepts including TCP socket creation, hostname resolution, IP address mapping, and bidirectional data exchange between networked applications.

2. EXPERIMENT OBJECTIVES

The primary objectives include gaining proficiency in socket programming concepts, understanding TCP socket creation using Python, mastering hostname resolution techniques, establishing reliable connections between client and server applications, and implementing bidirectional communication channels for data exchange.

3. TECHNICAL OVERVIEW

3.1 Socket Programming Fundamentals

A socket serves as an endpoint for sending or receiving data across a computer network. This experiment utilizes TCP sockets which provide reliable, connection-oriented communication suitable for chat applications where data integrity is paramount.

4. IMPLEMENTATION ANALYSIS

The server implementation represents the foundational component of our client-server architecture. The `ServerApp` class extends `CustomTkinter's CTK` class, encapsulating all socket management functionality within its methods.

7. RESULTS & INFERENCE

The experiment achieved all stated objectives successfully. Key achievements include: successful implementation of connection-oriented TCP sockets, proper handling of client-server lifecycle events, real-time message exchange with visual feedback, and extension to multi-client support using Python's threading module.

Key Screenshots:

AMRITA VISHWA VIDYAPEETHAM – CHENNAI CAMPUS		
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING		
23CSE302 – COMPUTER NETWORKS		
EXPERIMENT 2		
INTRODUCTION TO SOCKET PROGRAMMING USING PYTHON		
Name: Shaik Muzzammil		Max Marks (10)
Roll No.: CH.SC.U4CSE24041	Section: CSE-A	
Date of Exp: 22/06/2026	Date of Submission: 28/06/2026	
OBJECTIVES:		
1. Students will gain skill set in Socket Programming		
2. Students will understand how a basic socket can be created using Python, resolve the hostname of a website to its IP address, and also establish a connection to the website's server		

Figure: Page 1

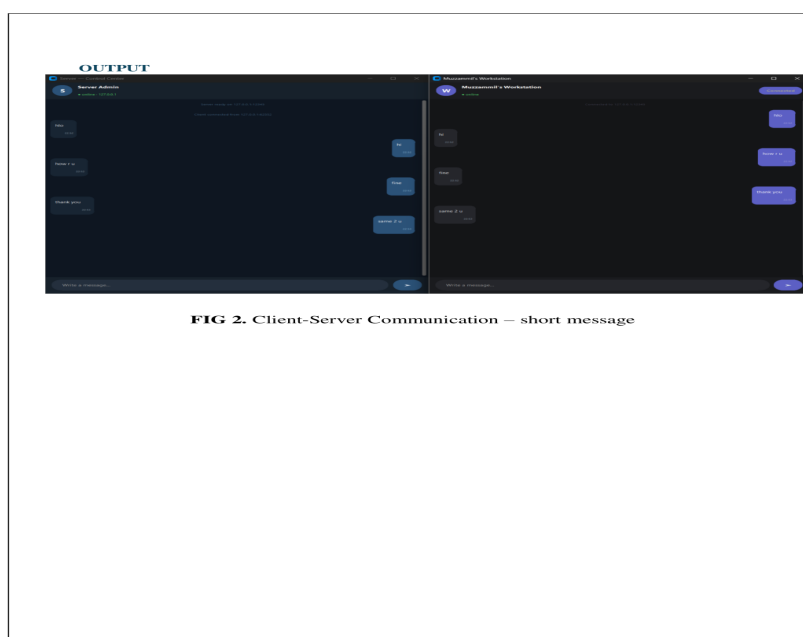


Figure: Page 10

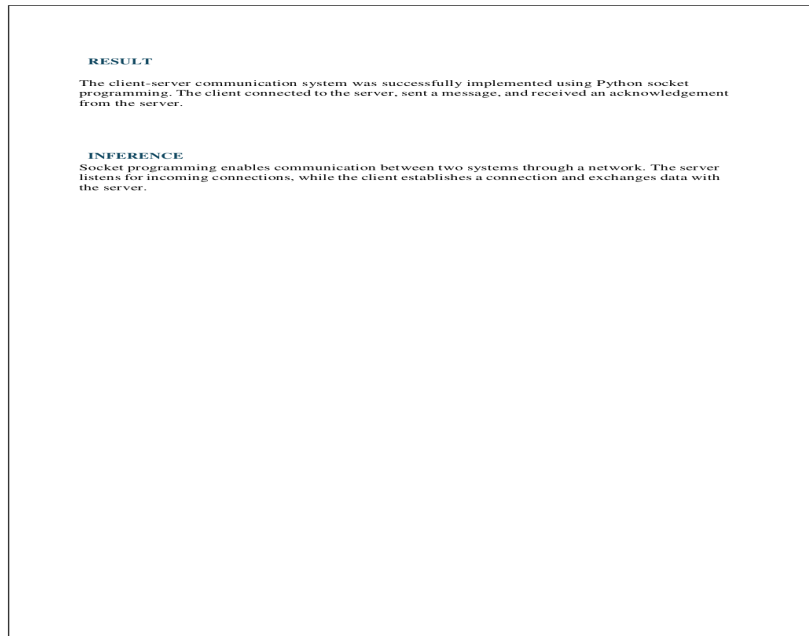


Figure: Page 11